

Nonsteroidal anti-inflammatory drugs

Introduction

- ❖ **Salicylates** and other similar agents used to **treat rheumatic disease**
- ✓ suppress the signs and symptoms of **inflammation & pain**. These drugs also exert antipyretic effects.
- ❖ Newer agents compared to Aspirin has
 - ✓ **Improved efficacy and decrease its toxicity.**

Chemistry & Pharmacokinetics

- ❖ All except nabumetone are weak organic acids
- ❖ Nabumetone is a **ketone** pro drug that is metabolized to the **acidic active** drug.

Chemistry & Pharmacokinetics

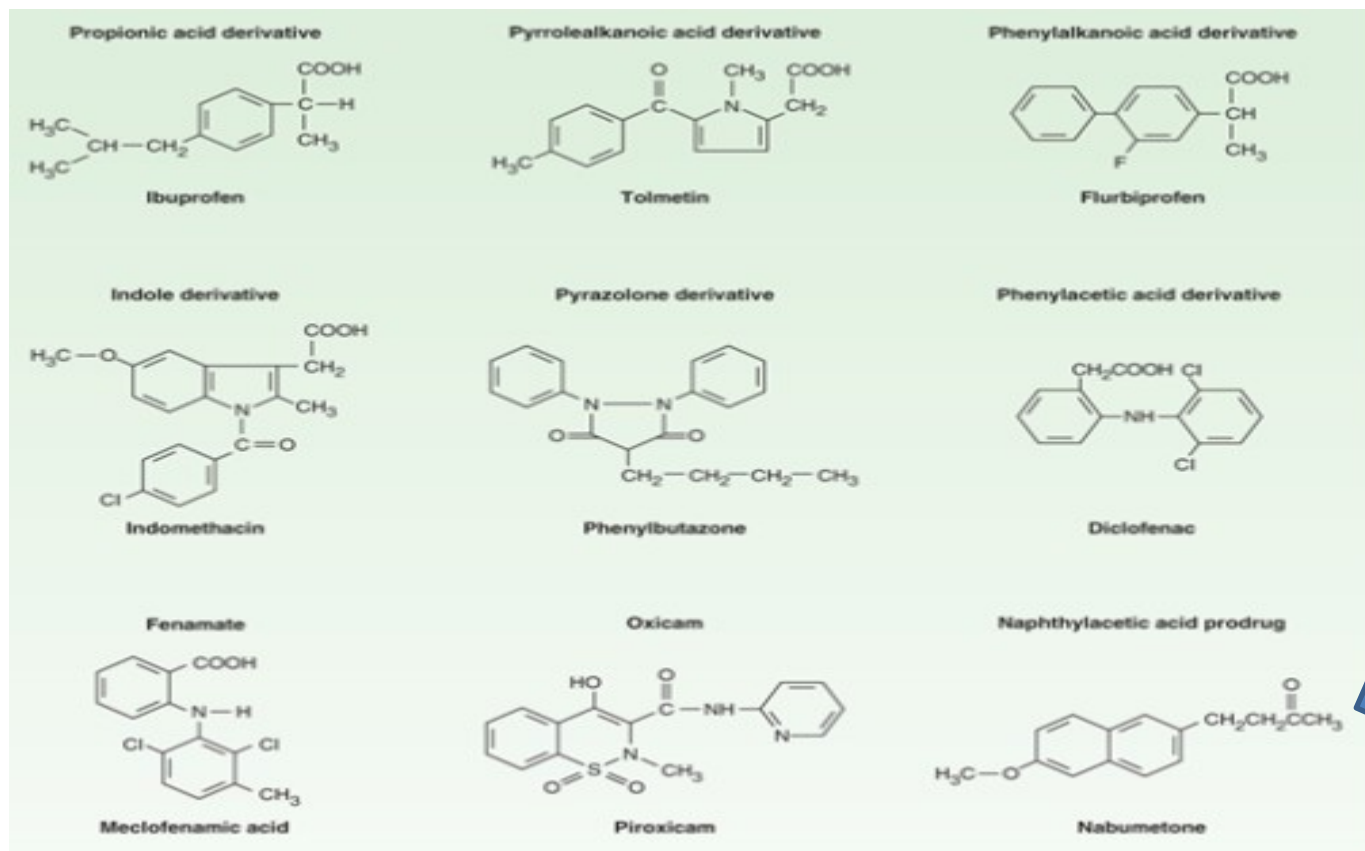


Figure: Chemical Structures of NSAIDs

Chemistry & Pharmacokinetics

❖ Most of are **well absorbed**, and **food does not** substantially **change their bioavailability**.

❖ **Metabolism ...Most by phase I** followed by **phase II**

Some direct glucuronidation **phase II** alone.

Excretion.....**renal excretion** is the most important route for **final elimination**,

Nearly all undergo varying degrees of biliary excretion and reabsorption (**enterohepatic circulation**).

✓ highly protein-bound (~ 98%), usually to albumin.

✓ Most of the NSAIDs (eg, ibuprofen, ketoprofen) are racemic mixtures,

Chemistry & Pharmacokinetics

- ❖ All NSAIDs can be found in synovial fluid after repeated dosing

Pharmacodynamics

Main mechanism: Inhibition of prostaglandin biosynthesis

- Aspirin irreversibly **acetylates** and **blocks platelet COX**, while the non-COX-selective NSAIDs are reversible inhibitors.

Other mechanisms:

- ✓ Inhibition of chemotaxis,
- ✓ down-regulation of IL-1 production,
- ✓ decreased production of free radicals and superoxide, and
- ✓ interference with calcium-mediated intracellular events.

Pharmacodynamics...

- ❖ selective **COX-2 inhibitors** do not affect platelet function
- ❖ Compared to older NSAIDs, COX-2-selective Inhibitors
 - ✓ Improved Efficacy
 - ✓ Less GI side effects
 - ✓ Can increase the incidence of edema, hypertension, and possibly, myocardial infarction
- ✓ NSAIDs decrease the sensitivity of vessels to bradykinin and **histamine**,
- ✓ affect lymphokine production from lymphocytes, and reverse the vasodilation of inflammation.

Pharmacodynamics...

- ✓ All newer NSAIDs have analgesic, anti-inflammatory, and antipyretic, and all (except the **COX-2–selective agents** and the **non acetylated salicylates**) inhibit **platelet aggregation**

- ❖ All NSAIDs have Nephrotoxicity,
 - ✓ Due to autoregulation of renal blood flow, which is modulated by prostaglandins

- ❖ Hepatotoxicity also can occur with any NSAID.

- ❖ Several NSAIDs (including aspirin)
 - ✓ Chronic use...reduce incidence of colon cancer
 - The mechanism for this protective effect is unclear.

Aspirin

- ❖ Aspirin's long use and availability without prescription
 - ✓ Diminishes its use compared to the newer NSAIDs.
 - ✓ Widely used for antiplatelet effects (i.e, doses of 81–325 mg once daily).

Pharmacokinetics

- ❖ Aspirin (acetylsalicylic acid; ASA) has a pKa of 3.5
- ❖ Aspirin is rapidly hydrolyzed (serum half-life 15 minutes) to **acetic acid** and **salicylate** by **esterases** in tissue and blood.
- ❖ Salicylate bound to albumin.
- ❖ **Alkalinization** of the urine increases the rate of excretion

Pharmacokinetics

Mechanisms of Action: Aspirin irreversibly inhibits platelet COX so that aspirin's antiplatelet effect lasts 8–10 days (the life of the platelet).

- ✓ In other tissues, synthesis of new **COX replaces** the **inactivated enzyme** so that ordinary doses have a duration of action of **6–12 hours**.
- ✓ **Clinical Uses:**
- ✓ (-)the incidence of transient **ischemic attacks**, **unstable angina**, coronary artery thrombosis with **MI** , and **thrombosis**

Adverse effects

❖ side effects of aspirin with antithrombotic dose

- ✓ gastric upset (intolerance)....Common

- ✓ Hepatotoxicity

- ✓ asthma,

- ✓ rashes,

- ✓ GI bleeding

} less common

❖ **C/I....**The antiplatelet action of aspirin is C/Is with hemophilia.

❖ Although previously not recommended during pregnancy, aspirin may be valuable in treating **preeclampsia-eclampsia**.

Nonacetylated salicylates

Drugs In this class..... magnesium choline salicylate, sodium salicylate, and salicyl salicylate. They have

- ✓ **Anti-inflammatory effect but**
- ✓ **Do not** inhibit platelet aggregation.
- ✓ They may be preferable when **COX inhibition** is undesirable such as in patients with **asthma**, those with **bleeding tendencies**, and even (under close supervision) those with **renal dysfunction**.
- ✓ The nonacetylated salicylates are administered in doses up to **3–4 g of salicylate** a day and can be monitored using serum **salicylate measurements**.

Cox-2 selective Inhibitors

COX-2 selective inhibitors, or **coxibs** **inhibit only** COX-2 isozyme induced at sites of inflammation

✓ COX-2 inhibitors do not **have impact on platelet aggregation,**

- Platelet aggregation is mediated by thromboxane produced by the COX-1 isozyme.

- Do not offer the cardioprotective

❖ Recommended doses of COX-2 inhibitors **cause renal toxicities** similar to traditional NSAIDs.

Celecoxib

- ❖ Celecoxib is a selective (**10–20** times more for COX-2)
 - ✓ Dose = **100–200 mg bid**
 - ✓ **T_{1/2}=11**
- ❖ S/E....cause rashes.
- ❖ It interacts occasionally with warfarin—as would be expected of a drug metabolized via CYP2C9.

Meloxicam

- ❖ Meloxicam is related to piroxicam
- ❖ preferentially inhibits COX-2 (but less than Celecoxib)
- ❖ Therapeutic dose of 7.5 mg/d.
- ❖ It is associated with fewer clinical GI symptoms and complications than piroxicam, diclofenac, and naproxen. Similarly
- ❖ Inhibit TXA₂, even at supratherapeutic doses,
 - ✓ does not reach result in (-)**platelet function**

Nonselective cox inhibitors

Drugs in this class: Diclofenac, Diflunisal, Etodolac, Flurbiprofen, Ibuprofen, Indomethacin, Ketoprofen, Nabumetone, Naproxen, Oxaprozin, Piroxicam, Sulindac, **Tolmetin**

Diclofenac

Dose: 50–75 mg qid

T_{1/2} = 1.1

S/E..... GI ulceration may occur less frequently than with some other NSAIDs.

- ✓ **Diclofenac/misoprostol** decreases upper GI ulceration but may result in diarrhea
- ✓ **Diclofenac/omeprazoleprevent of recurrent bleeding, but renal adverse effects** were common in high-risk patients.

Diclofenac, 150 mg/d,....impair renal blood flow and GFR..

A 0.1% ophthalmic preparation is promoted for prevention of postoperative ophthalmic inflammation

Diffunisal

Diffunisal is derived from salicylic acid but

- ✓ not metabolized to **salicylic acid** or **salicylate**.
- ✓ Active after It undergoes cleavage of the glucuronide moiety.
Diffunisal is subject to capacity-limited metabolism,
- ✓ In RA the recommended dose is **250-500 mg BID**.
- ✓ **T_{1/2}....13 hr**

Etodolac

- ✓ Etodolac is a racemic with 6.5 (t_{1/2})
- ✓ Dose...200–400 mg TID/QID times daily.
- ✓ The recommended dose in **OA** and **RA** 300 mg BID/TID---
500 mg twice

Flurbiprofen

- ❖ Flurbiprofen is a propionic acid derivative
- ❖ MOA...Its (S)(-) enantiomer inhibits COX non selectively, in addition
 - ✓ Flurbiprofen is affect tumor necrosis factor α (TNF- α) and
 - ✓ nitric oxide synthesis.
- ❖ a topical ophthalmic formulation used for inhibition of **intraoperative miosis**.
- ❖ flurbiprofen is also rarely associated with cogwheel rigidity, ataxia, tremor, and myoclonus.

Ibuprofen

- ❖ It is a simple derivative of phenylpropionic acid
- ❖ **Dose** = (<1600 mg/d)
- ❖ $T_{1/2}$2hr
- ❖ Compared to indomethacin, ibuprofen
 - ✓ Less decreasing Urine output effect
 - ✓ causes less fluid retention
- ❖ When given with aspirin
 - ✓ antagonizes the irreversible platelet inhibition induced by aspirin.
 - Limit aspirin's **cardioprotective effects**

Indomethacin

- ❖ It is an indole derivative with a potent inhibitor activity

Other mechanism

- ✓ may also inhibit phospholipase
 - ✓ reduce neutrophil migration, and
 - ✓ decrease T-cell and B-cell proliferation
- ❖ Dose: 50–70 mg tid
 - ❖ $T_{1/2}$4–5

Ketoprofen

- ❖ Ketoprofen is a propionic acid derivative that inhibits **both COX** (nonselectively) and **lipoxygenase**.
- ❖ Concurrent administration of probenecid elevates **ketoprofen levels** and prolongs **its plasma half-life**.
- ❖ Dosagr.....100–300 mg/d is equivalent to that of other NSAIDs.
- ❖ $T_{1/2}$ 1.8 hr

Nabumetone

- ❖ Nabumetone is the only nonacid NSAID in
 - ✓ it is given as a ketone prodrug
- ❖ . Its half-life > 24 hours
- ❖ Compared to other NSAIDs.....Less damaging GI effect
- ❖ higher dosages (eg, 1500–2000 mg/d)
- ❖ It is a very expensive NSAID.

Oxaprozin

- ❖ Oxaprozin is a propionic acid derivative NSAID.
- ❖ difference from the other members of this subgroup is a very long half-life (50–60 hours),
- ❖ It is **mildly uricosuric**
- ❖ the drug has the same benefits and risks that are associated with other NSAIDs.

Piroxicam

- ❖ It is an oxicam with Effects at **high concentrations**
 - ✓ inhibits polymorphonuclear leukocyte migration,
 - ✓ decreases oxygen radical production, and
 - ✓ inhibits lymphocyte function.
- ❖ $T_{1/2} = 57 \text{ hr}$
- ❖ Analgesic dose: 20 mg QD

Sulindac

- ❖ Sulindac is a sulfoxide prodrug.
- ❖ metabolized to the active sulfide metabolite and has **enterohepatic cycling**
 - prolongs the duration of action to 12–16 hours.

Uses

- ❖ RA, suppresses familial intestinal **polyposis** and it may inhibit the development of **colon, breast, and prostate cancer**

Sulindac..

- ❖ Among the more severe adverse reactions
 - ✓ Stevens-Johnson epidermal necrolysis syndrome,
 - ✓ thrombocytopenia,
 - ✓ agranulocytosis, and
 - ✓ nephrotic syndrome
- cholesteric liver damage (some times)

Drugs used for Gout

Introduction

- ❖ Gout is a metabolic disease characterized by
 - ✓ recurrent episodes of acute arthritis due to deposits of **monosodium urate** in **joints and cartilage**.
- ❖ Gout is usually associated with a high **serum uric acid** level (**hyperuricemia**),
- ❖ a poorly soluble substance that is the major end product of purine metabolism.
 - ✓ In most mammals, **uricase** converts uric acid to the more soluble **allantoin**;
 - This enzyme is absent in humans.

Introduction

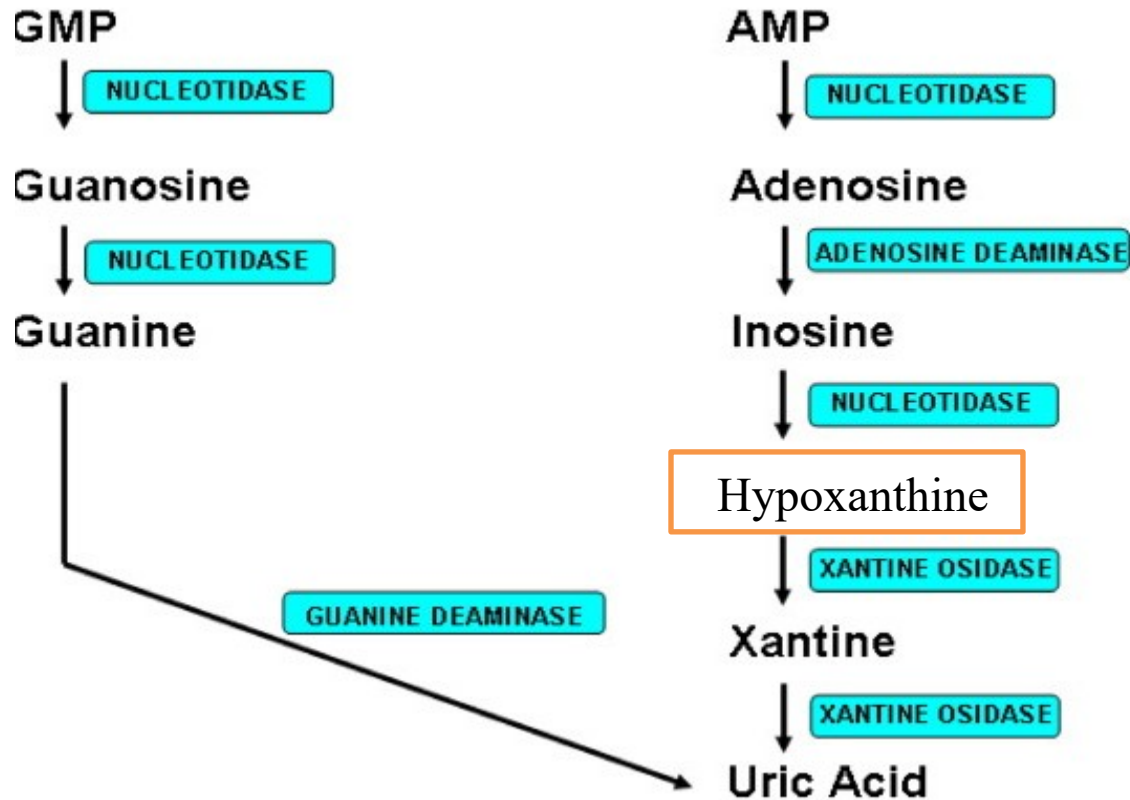


Figure: Enzymatic degradation of purines in humans.

Introduction

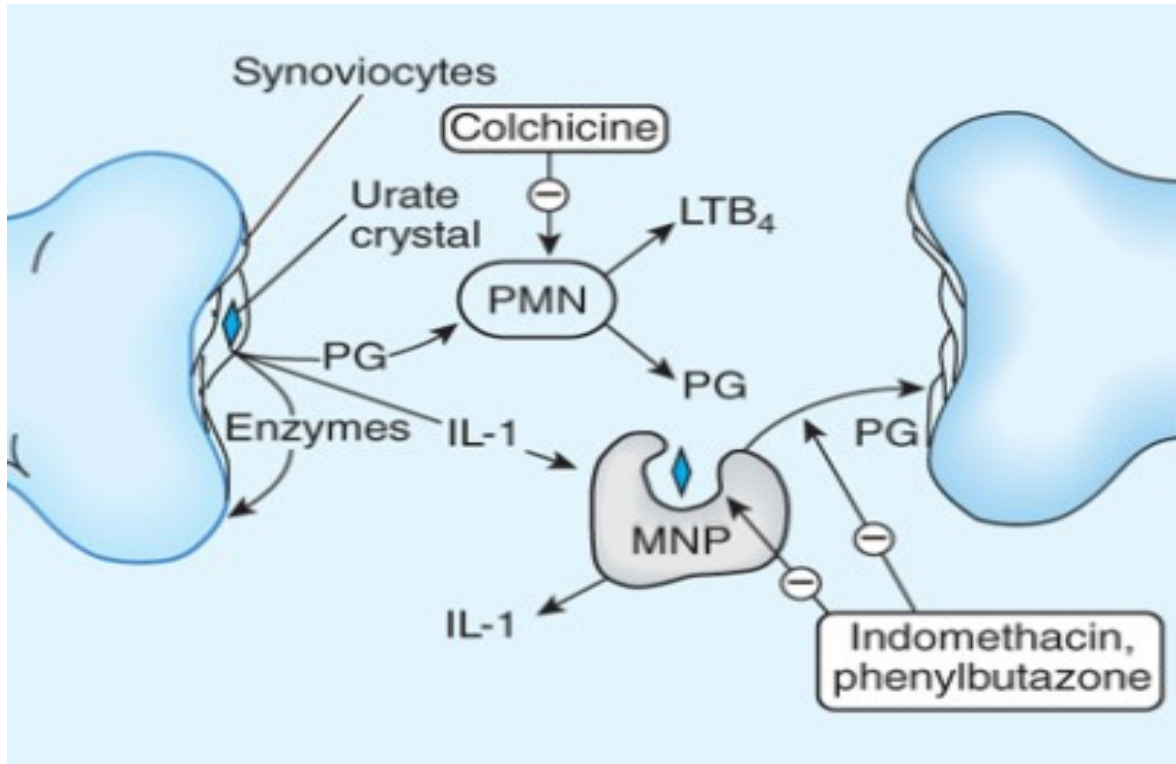
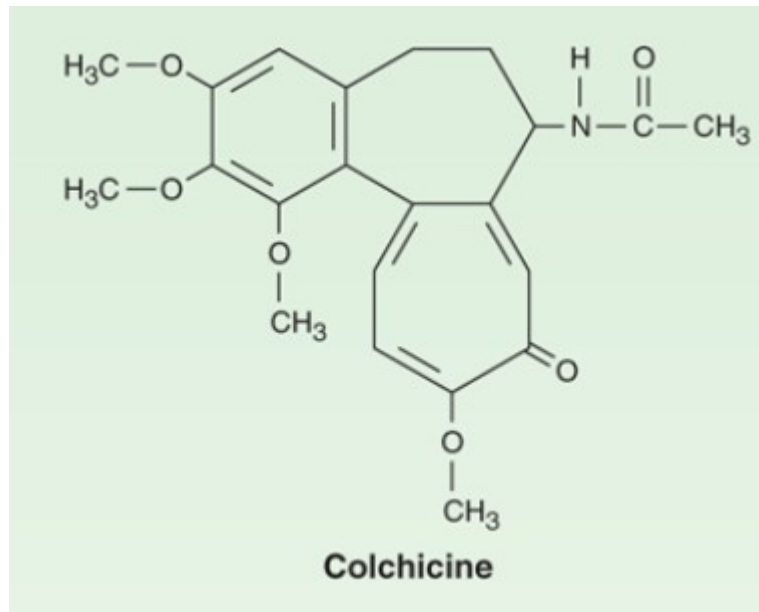


Figure: Pathophysiologic events in a gouty joint

Drugs

- **Colchicine**

It is an alkaloid isolated from the autumn *crocus*, *Colchicum autumnale*.



Pharmacokinetics

- ❖ Colchicine after oral admin
- ❖ Peak plasma levels within 2 hours,
- ❖ serum half-life = 9 hours.

Pharmacodynamics: Colchicine relieves the pain and inflammation of gouty arthritis **in 12–24**

- ❖ Do not have an effect on
- ✓ metabolism or excretion of urates and is **without other analgesic** effects.

Pharmacodynamics

MOA..... anti-inflammatory effect

- ✓ bind to the intracellular protein tubulin
- ↓
- ✓ preventing its polymerization into microtubules
- ↓
- ✓ inhibition of leukocyte migration and phagocytosis.
- ❖ It also inhibits the formation of leukotriene **B4** and **IL-1 β** .

Indications

- ❖ indicated for gout and is also used between attacks (the “intercritical period”) for prolonged prophylaxis (at low doses).

Dose: 0.6 mg one to **three** times daily (prevention)

1.2 mg followed by a single 0.6-mg (Treatment of attack)

- ❖ It prevents attacks of acute **Mediterranean fever** and may have a mild beneficial effect in **sarcoid arthritis** and in **hepatic cirrhosis**.
- Colchicine is also used to treat and prevent **pericarditis**, **pleurisy**, and **coronary artery**

colchicine

- ❖ Adverse Effects: Colchicine often causes.....**diarrhea**
- ❖ occasionally
 - ✓ nausea,
 - ✓ vomiting, and
 - ✓ abdominal pain.
 - ✓ Hepatic necrosis,
 - ✓ acute renal failure seizures
- ❖ Rarely... it cause **hair loss** and **bone marrow depression**, peripheral neuritis, myopathy, and, in some cases, **death**.
- ❖ The more severe adverse events have been associated with the **IV administration** of colchicine.

NSAIDs In Gout

- ❖ NSAIDs inhibit urate crystal phagocytosis.
- ❖ Aspirin is not usedit causes renal retention of uric acid at **low doses (≤ 2.6 g/d)**.
- ❖ Aspirin should not be given for analgesia
It is uricosuric at doses greater **than 3.6 g/d**.
- ✓ **Indomethacin**used in the initial treatment of gout as a replacement for colchicine.
- ✓ For acute gout, 50 mg (TID) after a response the dosage is 25 mg **TID for 5–7 days**.

Uricosuric agents

Drugs in this class: Probenecid and sulfinpyrazone

- ✓ with **tophaceous gout** or in those with increasingly frequent gouty attacks.
- ✓ **Newer agent.... Lesinurad** .is a promising new uricosuric agent that is currently in phase 3 trials.

Dose

- ✓ Probenecid Initially **0.5 g** orally daily in divided doses -**1 g daily after 1 week**.
- ✓ Sulfinpyrazone started (**200 mg** orally daily, progressing to **400–800 mg** daily).

Pharmacodynamics

Uricosuric drugs

- inhibit active transport sites for **reabsorption** in the **proximal renal tubule**

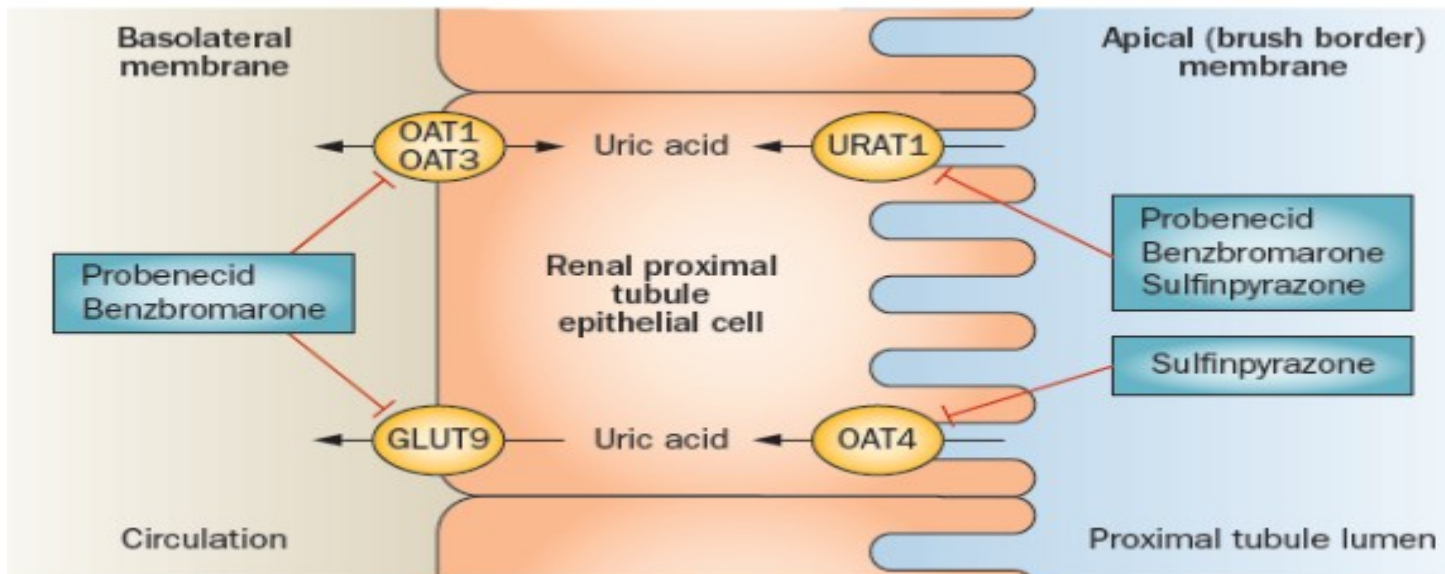


Figure: Mechanism of Action

Uricosuric agents....

Indications: Uricosuric therapy should be initiated in gouty patients with

- ✓ under excretion of uric acid
- ✓ when allopurinol or febuxostat is C/I or when
- ✓ tophi are present.

Adverse Effects:

- ✓ GI irritation,
- ✓ rarely cause aplastic anemia.
- ✓ A rash can be developed
- ✓ Nephrotic syndrome (with [probenecid](#))

Allopurinol

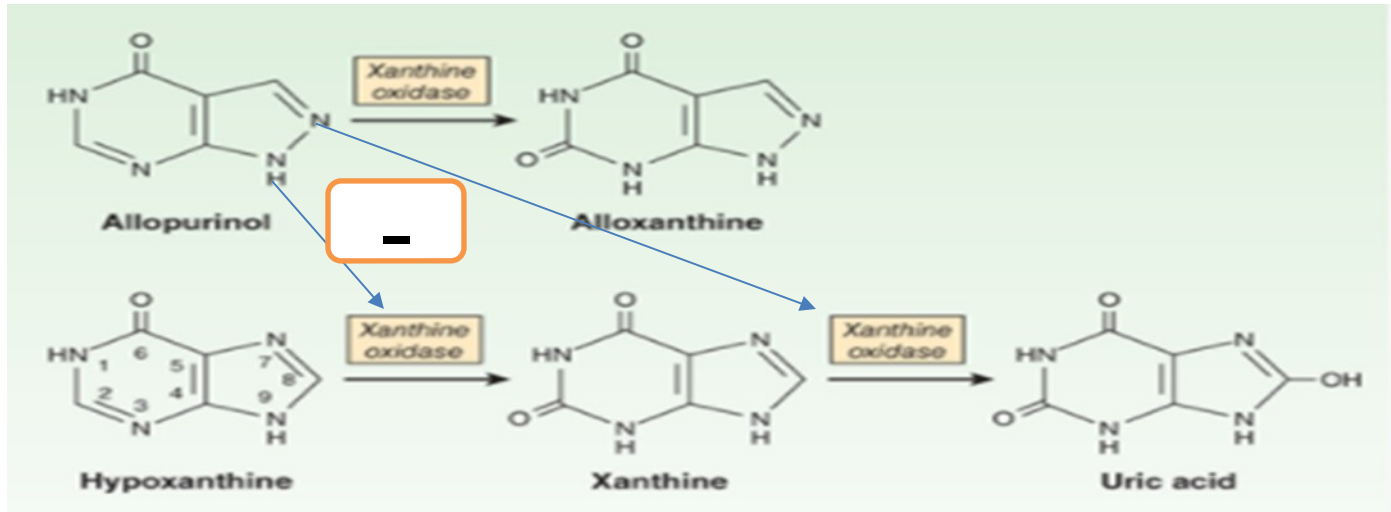
- ❖ **XO inhibitor drugs.....**Allopurinol, topiroxostat and Febuxostat
- ❖ **Allopurinol....**It inhibiting xanthine oxidase and thereby formations of uric acid

Chemistry and Pharmacokinetics

- ❖ Absorption...80% and terminal
- ❖ half-life of **1–2** hours
- ❖ allopurinol is metabolized by xanthine oxidase to produce, **alloxanthine** (has an effect in – uric acid formation)
 - **allopurinol once a daily might be suffice**

Mechanisms of Actions

MOA: Allopurinol competitively inhibits XO at **low concentrations** and is a noncompetitive inhibitor at **high concentrations**



Indications

Allopurinol is often the first-line

- ✓ For treatment of **chronic gout** in the period between **attacks** and it tends to prolong the **intercritical period**.
 - ✓ When initiating allopurinol, colchicine or NSAID should be used until steady-state, Uric acid (<6 mg/dL) should be continued for **6 months** or >
- 
- ✓ **D/c** colchicine or the NSAID while continuing allopurinol therapy.

Dosage

- ❖ For $GFR > 40$ ml/min
- ❖ 100 mg initially...increase weekly by **100-mg**
- ❖ Most patients can be maintained on 300 mg/day.
- ❖ 300 mg/day
- ❖ With frequent monitoring for adverse effects, daily doses may exceed 300 mg, even in renal impairment.
- ❖ Individuals with **hematological malignancies**
 - may need up to 800 mg/day beginning 2 to 3 days before the start of chemotherapy..

Adverse Effects:

S/E of Allopurinol

- precipitating gout
- GI intolerance (including nausea, vomiting, and diarrhea),
- peripheral neuritis
- necrotizing vasculitis,
- bone marrow suppression and
- aplastic anemia may rarely occur.
- An allergic skin lesions (3%)

Interactions and Cautions

- ❖ Allopurinol +azathioprine
 - ✓ Dose of azathioprine should be ↓ by 75%.
- ❖ Allopurinol (++) S/E of cyclophosphamide.
- ❖ Allopurinol (--) the metabolism of **probenecid** and oral **anticoagulants** and may increase **hepatic iron concentration**.
- ❖ Safety in **children** and **during** pregnancy has not been established

Febuxostat

- ❖ is a non-purine xanthine oxidase inhibitor that was approved by the FDA in 2009.
- ❖ Pharmacokinetics:
 - ✓ more than 80% absorbed
 - ✓ Max conce.....in 1 hour and a
 - ✓ half-life of 4–18 hours, **once-daily dosing is effective**

Pharmacodynamics

❖ Febuxostat is a potent and selective inhibitor of xanthine oxidase,.

❖ **Indications:** Febuxostat is approved for chronic hyperuricemia

Dose40 or 80 mg

Adverse effects

- ❖ Treatment with **colchicine** or **NSAIDs** should be startedto **avoid gout flares**.
- ❖ The most frequent treatment-related adverse events are
 - ✓ liver function abnormalities,
 - ✓ diarrhea,
 - ✓ headache, and nausea.
- ❖ **Febuxostat** is well tolerated in patients with a history of **allopurinol intolerance**.

Pegloticase

- ❖ Pegloticase is the **newest urate-lowering** therapy to be approved for the treatment of refractory chronic gout
- ❖ Chemistry: Pegloticase is a recombinant mammalian uricase that is covalently attached to methoxy polyethylene glycol (mPEG)
 - to prolong the circulating half-life and diminish immunogenic response.

Dose

- ❖ 8 mg every 2 weeks administered as an intravenous infusion.
- ❖ It is a **rapidly acting** drug, achieving a peak decline in uric acid level within **24–72 hours**.
- ❖ The serum half-life 6 to 14

Pharmacodynamics

- ❖ **Urate oxidase enzyme**, absent in humans and some higher primates
 - ✓ converts uric acid to **allantoin**.
 - ✓ is highly soluble and can be easily eliminated by the **kidney**.
- ✓ MOA.... catalyzing oxidation of **uric acid** to allantoin

Adverse Effects

- ❖ **Gout flare (with first 3–6 months of treatment)**
 - requiring prophylaxis with **NSAIDs** or **colchicine**.
 - Large numbers of **patients show** immune responses to **pegloticase**.
- ✓ Infusion reactions and anaphylaxis (6–15% of patients receiving pegloticase).

Adverse Effects

less frequent side effects

- ✓ upper respiratory tract infection, peripheral edema, urinary tract infection, and diarrhea.
- ✓ **hemolytic anemia** in ...G6PDH deficiency
 - formation of **hydrogen peroxide** by uricase; therefore,
 - pegloticase should be avoided in these patients.

Glucocorticoids

- ❖ **Corticosteroids** are used in the treatment of severe symptomatic gout, by **intra-articular** or **SC**,
——— depending on the degree of pain and inflammation.
- ❖ The most commonly used oral corticosteroid is **prednisone**.
 - ✓ Dose = 30–50 mg/d for **1–2 days**, tapered **over 7–10 days**.
- ❖ Intra-articular injection of **triamcinolone acetonide**
 - ✓ if the patient is unable to take oral medications.

Interleukin 1 inhibitors

- ❖ Drugs targeting IL-1 for treatment of gout are
 - ✓ anakinra,
 - ✓ canakinumab, and
 - ✓ rilonacept, are for the treatment of gout.
- ✓ Used for acute gout in patients with contraindications to **NSAIDs** or **colchicine**..